

CAUSAL INFERENCE & MACHINE LEARNING

Evaluating Job Training Impact: LaLonde Dataset

Uncovering the True Effect on 1978 Earnings

Chetna Lourembam

The Policy Question

1

The Goal: Determine if a government-sponsored job training program genuinely increased participants' future earnings (in 1978).

2

The Problem: We are working with observational data, not a randomized controlled trial (RCT).

3

The Trap Selection Bias: People who choose to enter job training often have very different starting conditions (like lower prior incomes) than those who do not.

RCT

Random assignment removes confounding by design.

OBSERVATIONAL DATA (this project)

Treated and control groups self-select — confounders must be modeled, not assumed away.

The Data Science Approach

1

Standard Machine Learning is great at predicting outcomes, but it fails to answer causal questions ("Did X cause Y?").

2

Our Solution: We transition from classical predictive models to a rigorous Causal Inference pipeline.

Advanced weighting, matching, and machine learning (Double ML and Causal Forests) mathematically simulate a randomized trial isolating the true monetary impact.

Predictive ML

Great at forecasting Y from X



Causal Inference

Isolates the true effect of X on Y



DML + Causal Forests

Simulated randomization

Standing on the Shoulders of Giants

1986

LaLonde

The foundational paper for this dataset famously proving that traditional econometric models often fail to replicate experimental results when faced with severe selection bias.

2018

Chernozhukov et al.

Pioneered Double Machine Learning (DML), allowing us to use flexible ML algorithms to “partial out” confounding variables.

2018

Wager & Athey

Developed the Causal Forest, giving us the power to move beyond average effects and discover precisely who benefits the most from an intervention.

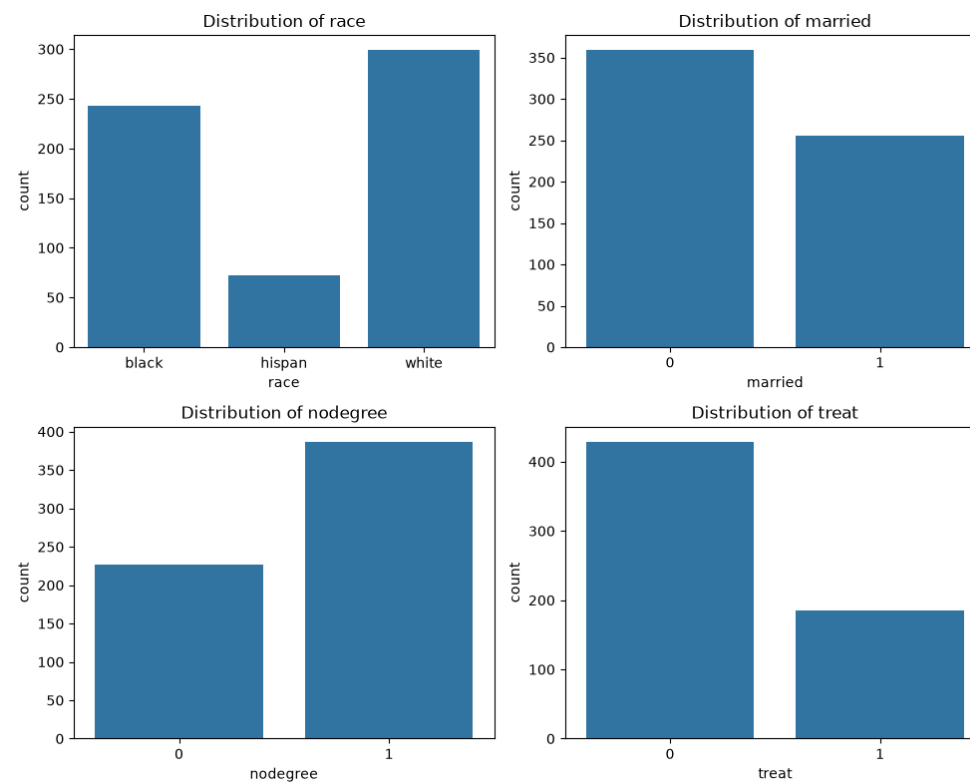
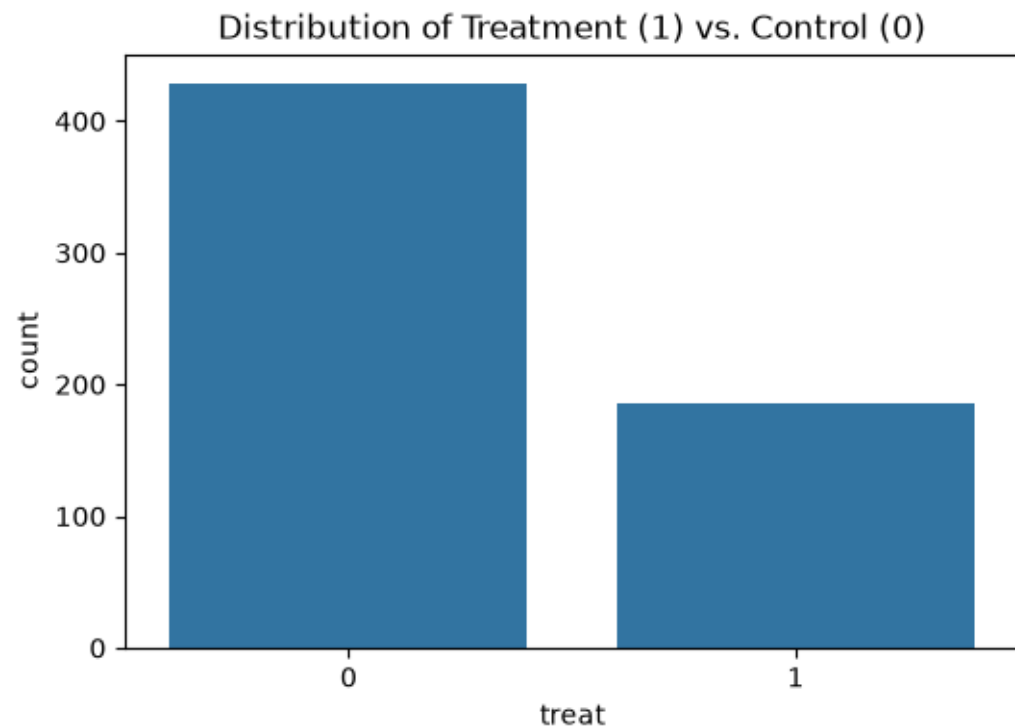
Initial Data Imbalance

1

The LaLonde dataset features severe class imbalance. We have far more individuals in the control group than the treated group.

2

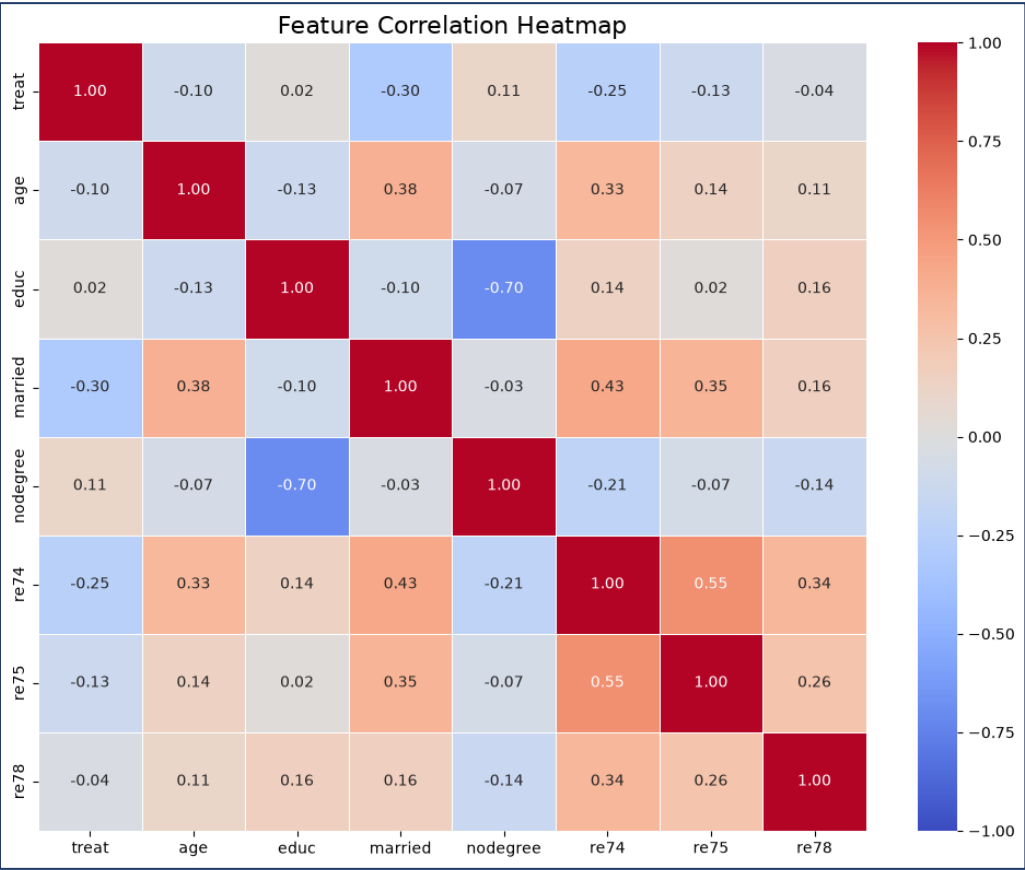
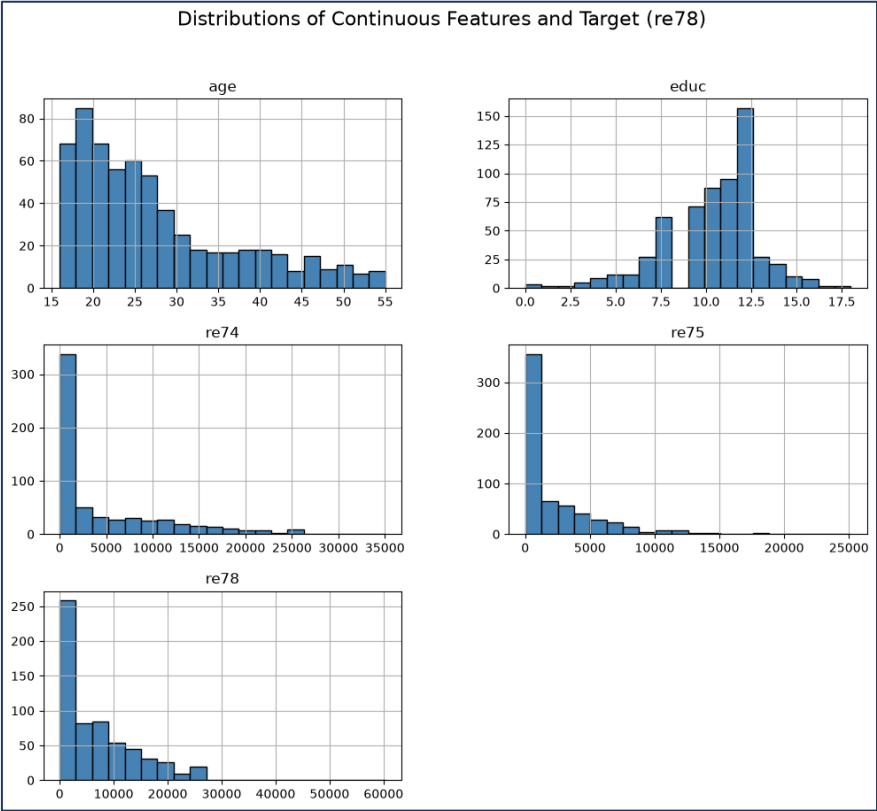
Looking at the demographics, the treatment group and control group have drastically different racial and marital distributions.



Income Skew and Feature Correlation

Historical earnings (re74 and re75) are heavily right-skewed, showing that a majority of our sample started near zero income.

The correlation heatmap reveals that being married and having a degree are negatively correlated with receiving treatment confirming our selection bias hypothesis.



The Naive Baseline (OLS)

MODEL

Ordinary Least Squares (OLS) Linear Regression with covariates.

FINDING

The naive regression estimates a massively inflated Average Treatment Effect (ATE).

WHY IT FAILS

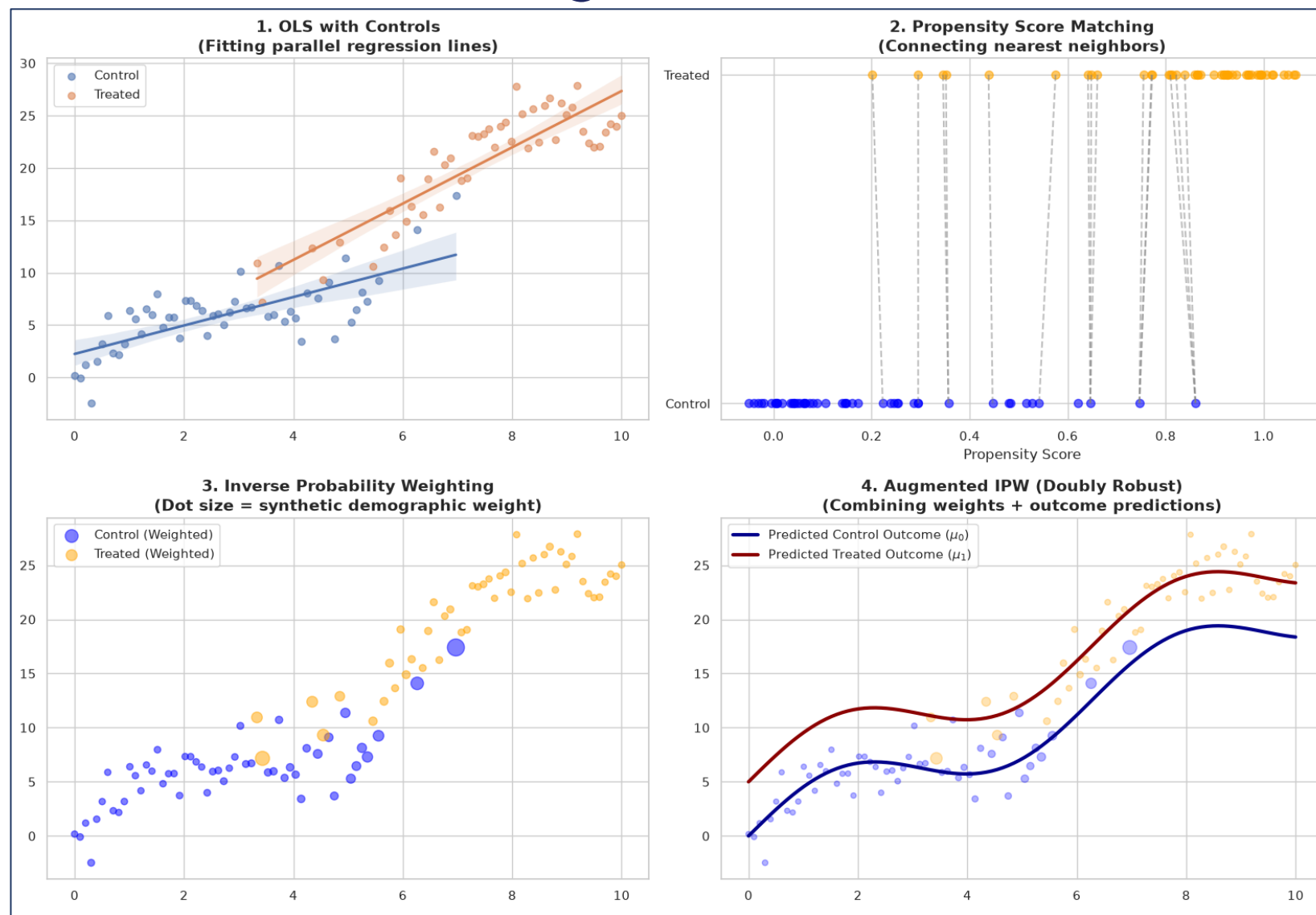
Standard regression assumes linear relationships and struggles to mathematically adjust for the fact that the treated and control groups occupy completely different historical income brackets.

NAIVE ATE ESTIMATE

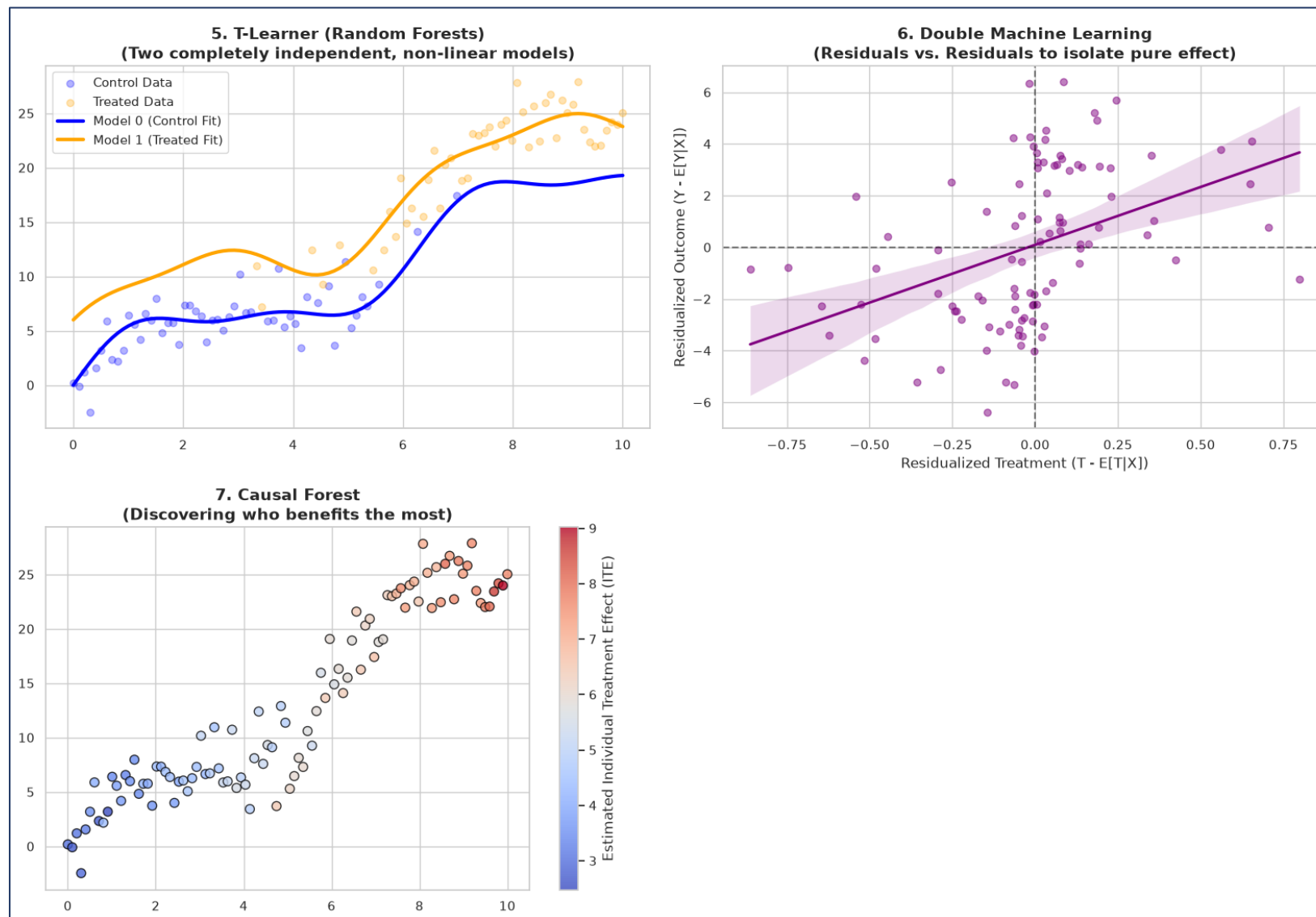
\$1,300+

*Overestimated relative to the causal-ML result
— a warning sign that confounding is
unaddressed.*

The Causal Machine Learning Toolkit



The Causal Machine Learning Toolkit



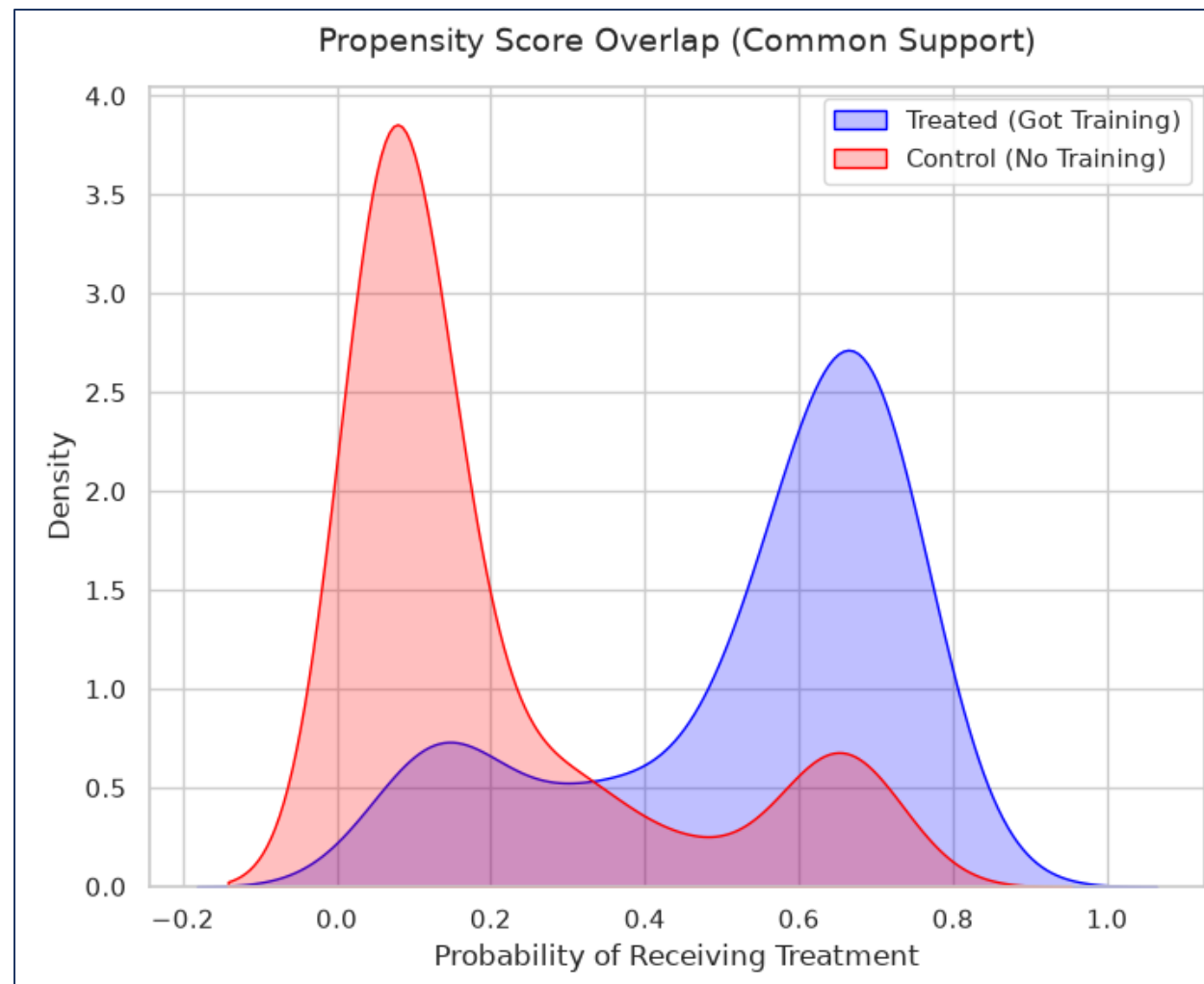
Identifying the Bias (Common Support)

1

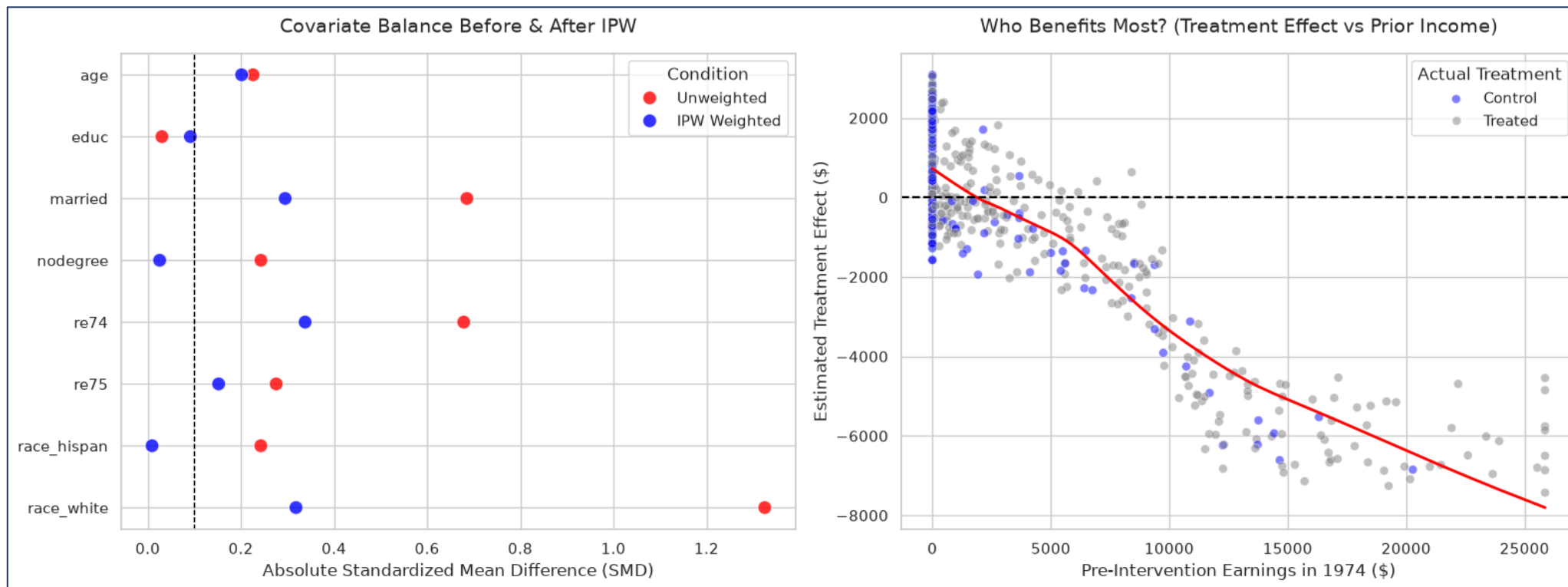
Before calculating outcomes, we must calculate the Propensity Score (probability of receiving training).

2

The density plot proves why advanced models are needed: the control group (red) had a massive spike near 0% probability of getting training, while the treated group (blue) clustered much higher.



Proving Model Validity (The Love Plot)



- ☐ Did our models actually fix the bias? Yes.
- ☐ This "Love Plot" measures the Standardized Mean Difference (SMD) of our demographics.
- ☐ Before weighting (red dots) the groups were wildly different. After applying Inverse Probability Weighting (blue dots), we successfully collapsed the bias below the 0.1 threshold effectively simulating a randomized trial.

Average Treatment Effect (ATE) Stabilization

1

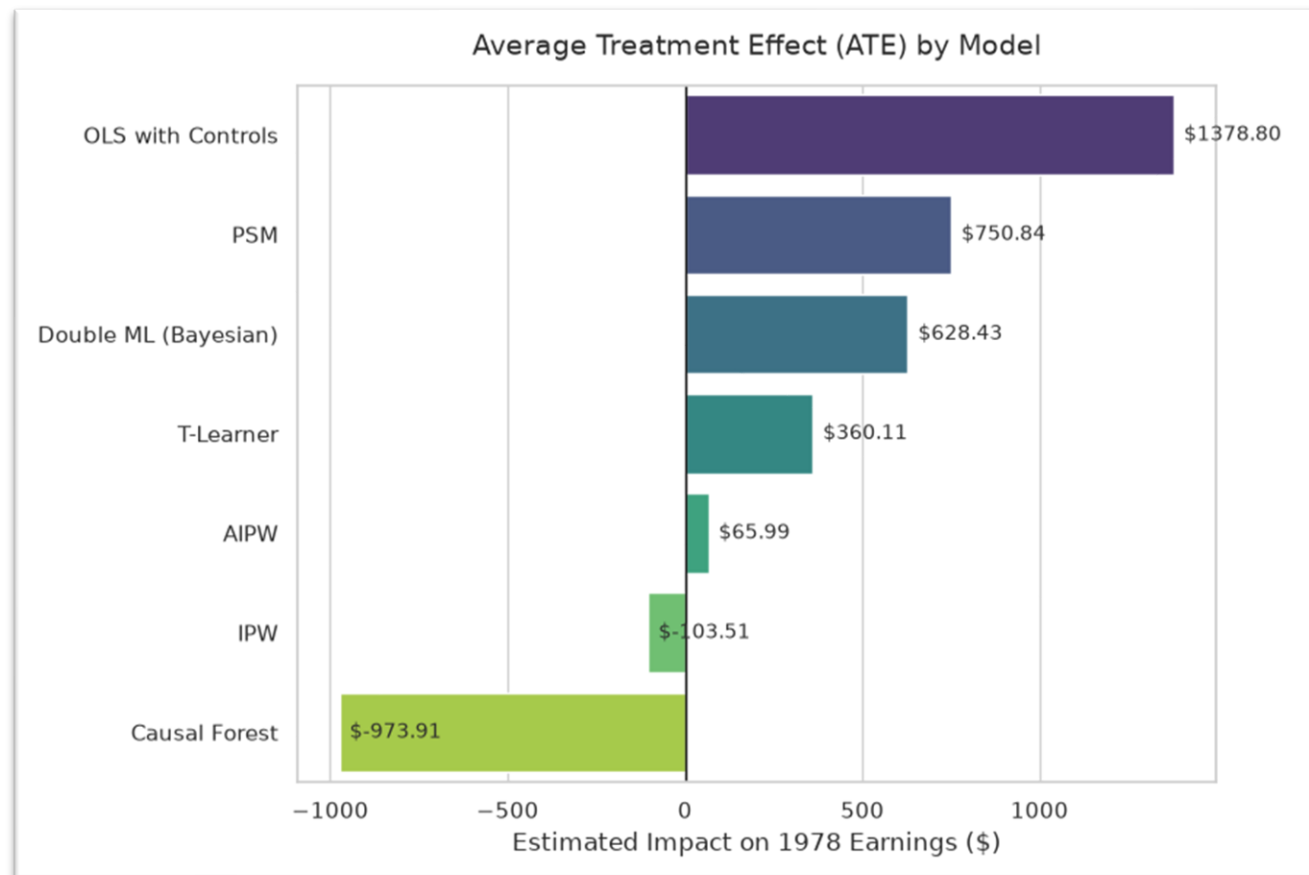
As our algorithms became better at filtering out demographic noise, our estimated monetary impact stabilized.

2

The naive OLS model overestimated the impact at over \$1,300.

3

Our advanced ML models (Double ML and Causal Forests) successfully adjusted for the historical poverty bias, bringing the true ATE down to a realistic, slightly negative/neutral overall average.



Heterogeneous Treatment Effects (ITE)

1

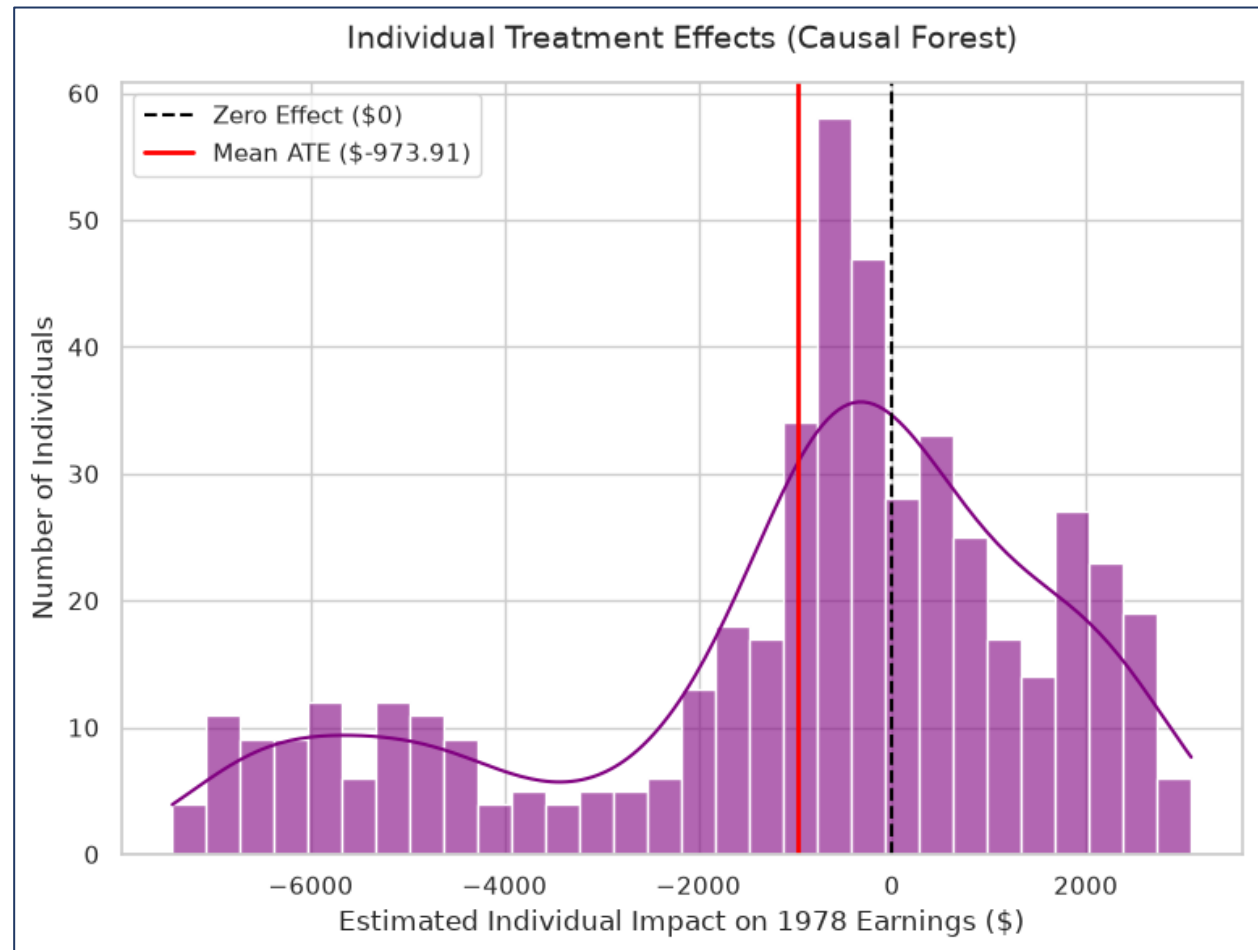
An average doesn't tell the whole story. The Causal Forest calculates the Individual Treatment Effect (ITE).

2

The distribution shows that while the average hovers near zero, there is a spread — some people lost ground, but a specific subset saw major positive gains.

3

Feature importance reveals that historical earnings (re74, re75) were the ultimate deciding factors in whether the program worked.



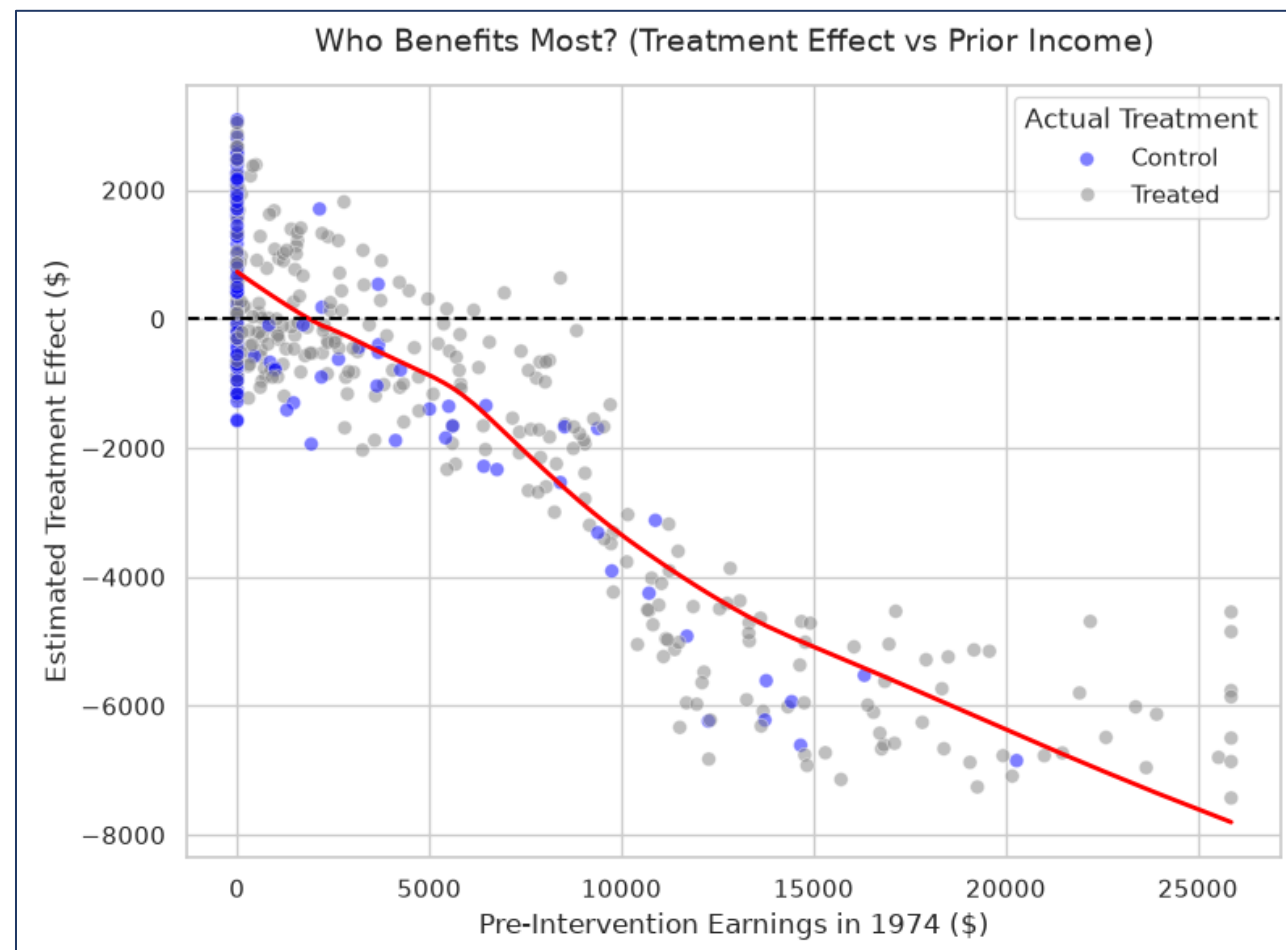
Who Benefits the Most?

1

Plotting individual treatment effects against pre-intervention earnings (1974) reveals a powerful policy insight.

2

The downward-sloping trendline indicates that the job training program yields the highest financial returns for individuals who started in the most extreme poverty.



Technical Hurdles & Resolutions

Dependency Hell

PROBLEM

Severe version clashing between econml, modern numpy, and numba.

RESOLUTION

Force-downgraded to a stable NumPy backend and used notebook-specific %pip commands.

Algorithm Strictness

PROBLEM

LinearDML initially crashed — it expected a continuous treatment variable instead of our binary (0/1) training indicator.

RESOLUTION

Swapped to a Logistic Regression first stage and explicitly set `discrete_treatment=True`.

Data Scarcity

PROBLEM

A relatively small sample size limits the depth of deep learning architectures.

RESOLUTION

Validated our choice of Random Forests and Ridge regressions instead.

Predictive vs. Causal Machine Learning

1

This project highlights the danger of trusting raw predictive ML for policy decisions.

2

Causal models force us to ask why the data looks the way it does before predicting an outcome.

“

If we had only used standard regression or a basic Random Forest, we would have told stakeholders the program was wildly successful for everyone — entirely due to omitted variable bias.

Real-World Policy Implications

1

The job training program is not a universal “silver bullet” for unemployment.

2

However, it is a highly effective intervention for a very specific demographic: individuals with little to no prior income.

RECOMMENDATION

Future funding should strictly target low-income brackets rather than general enrollment maximizing the ROI of government spending.

Conclusion & Future Work

By utilizing Double Machine Learning and Causal Forests, we successfully stripped away severe selection bias proving that causal ML is essential for observational data analysis.

01

FUTURE WORK

Replicate the Bayesian Causal Forest (BCF) using an R environment wrapper to generate more rigorous uncertainty intervals.

02

FUTURE WORK

Acquire longitudinal data to see if the treatment effect compounds 5, 10, or 20 years after the 1978 cutoff.